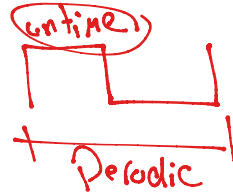
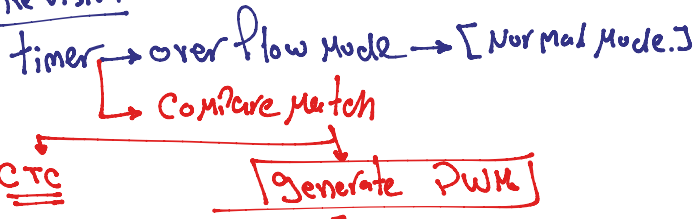


timer 2

$\frac{\text{ontime}}{\text{Periodic}} \times 100$



* Revision



PWM (Pulse width Modulation)

* why Need PWM?

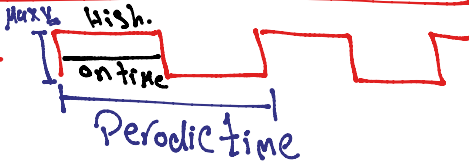
Control Intensity (led) Control speed Motor

MC output (digital) (1, 0)

to control on the output devices (analog)

* Solution
 ① search -> MC have DAC
 ② use PWM to control

Definition for PWM



- ① Periodic time
- ② Frequency
- ③ ontime -> High
- ④ off time -> low
- ⑤ Duty cycle -> Ratio between ontime / Periodic time. $\Rightarrow \frac{\text{on}}{\text{Pera}} \times 100$

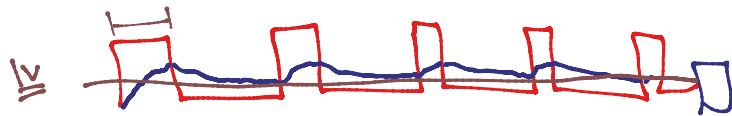
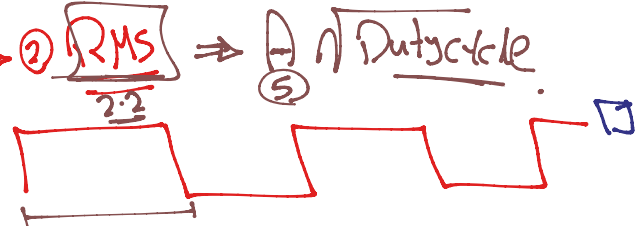
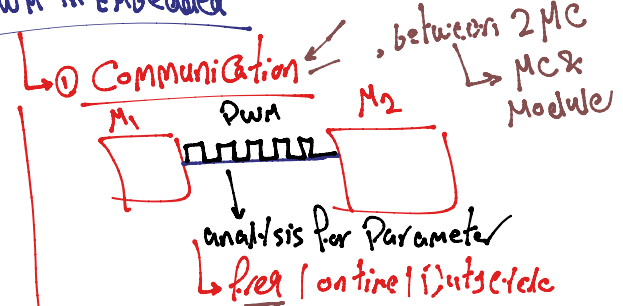
⑥ Amplitude

⑦ RMS -> (Root mean squared)

$$RMS = \text{Amplitude} \times \sqrt{\text{duty cycle}}$$

Motor 3.3V

PWM in Embedded

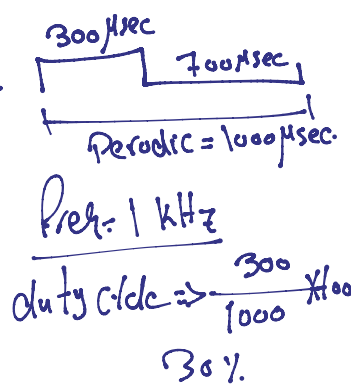


- ① PWM -> signal generated for mcu.
- ② PWM -> normal pulse control width

How generate PWM?

① by SW -> using delay

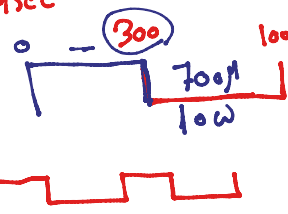
```
while(1)
{
    Motor on();
    delay_usec(300);
    Motor off();
    delay_usec(700);
}
```



② SW using (normal or CTC mode)

ISR(ovf) -> each 10µsec

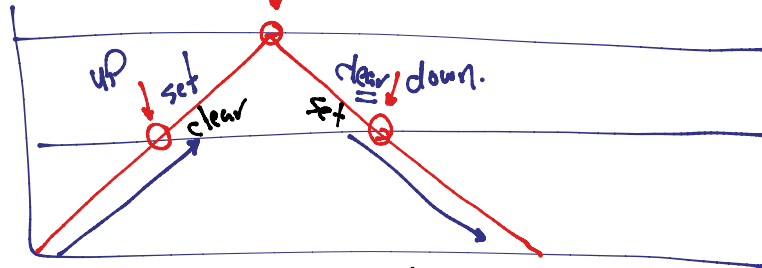
```
static uint16_t counter = 0;
counter++;
if (counter == 30)
{
    Motor off();
}
else if (counter == 100)
{
    Motor on();
    counter = 0;
}
```



Phase Correct

* → PWM Phase Correct (time & Presb)
 (up Count / down Count)

Action → up flag
 down flag



* Compare value = 2^n (Duty cycle) → Non Invert.
 clear / up
 set / down.

* Compare value = 2^n (1 - Duty cycle) → Invert
 clear / down
 set / up.

